Container With Most Water (Medium)

Problem Statement

Given an array of non-negative integers, where each integer represents the `height` of a vertical line positioned at index `i`. You need to find the two lines that, when combined with the x-axis, form a container that can hold the most water.

The goal is to find the maximum amount of water (area) that this container can hold.

Note: The water container's width is the distance between the two lines, and its height is determined by the shorter of the two lines.

Examples

Example 1:

- **Input**: [1,3,2,4,5]
- **Expected Output**: 9
- **Justification**: The lines at index 1 and 4 form the container with the most water. The width is 3 * (4-1), and the height is determined by the shorter line, which is 3. Thus, the area is 3 * 3 = 9.

Example 2:

- **Input**: [5,2,4,2,6,3]
- **Expected Output**: 20
- **Justification**: The lines at index 0 and 4 form the container with the most water. The width is 5 * (4-0), and the height is determined by the shorter line, which is 5. Thus, the area is 5 * 4 = 20.

Example 3:

- **Input**: [2,3,4,5,18,17,6]
- **Expected Output**: 17
- **Justification**: The lines at index 4 and 5 form the container with the most water. The width is 17 * (5-4), and the height is determined by the shorter line, which is 17. Thus, the area is 17 * 1 = 17.

Try it yourself

Try solving this question here:

Java

```java
public class Solution {
    // Function to compute the maximum area
    public int maxArea(int[] height) {
        int maxArea = 0; // To store the maximum area found
        // ToDo: Write Your Code Here.
        return maxArea;
    }
}
```

Test Case | Run Code Result | Stdout

[1,3,2,4,5]

Use Example Testcases | Run | Submit

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Solution: Maximum Product Subarray

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Solution: Container With Most Water

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